

07.- Docker Compose

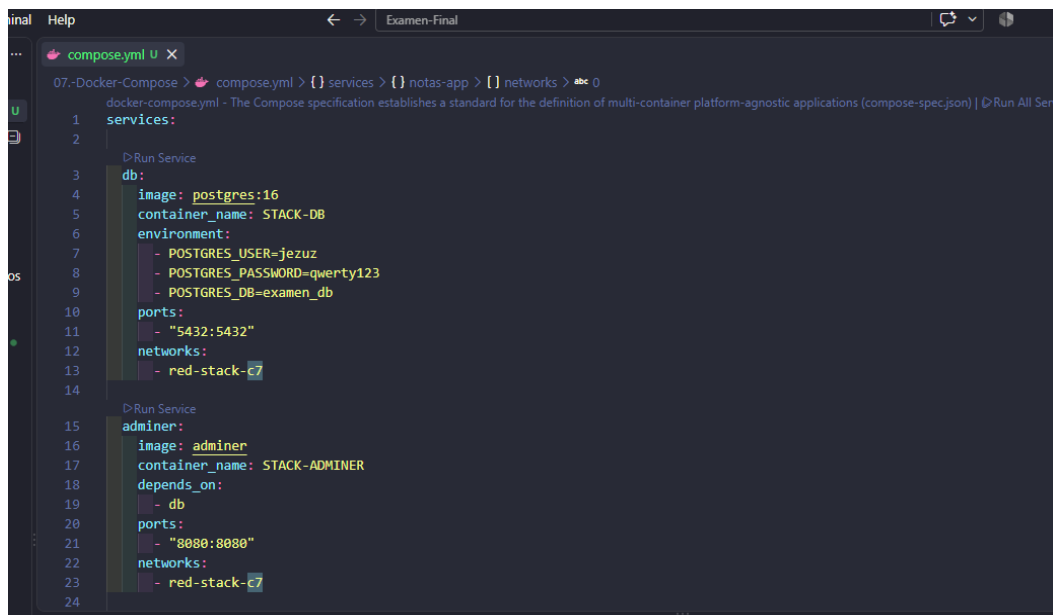
En esta actividad crearemos **5 contenedores utilizando Docker Compose**, a partir de las imágenes que hemos venido trabajando:

- postgres:16
- adminer
- redis:7
- jezuz1904/to-do-list-v1:1.0
- jezuz1904/tarjeta-notas-v2:1.0

Creación del archivo Compose

Comenzamos creando el archivo `compose.yml`. En este archivo definiremos toda la configuración necesaria para nuestros contenedores, incluyendo:

- Imágenes que se utilizarán.
- Puertos que serán expuestos.
- Red que permitirá la comunicación entre los contenedores.

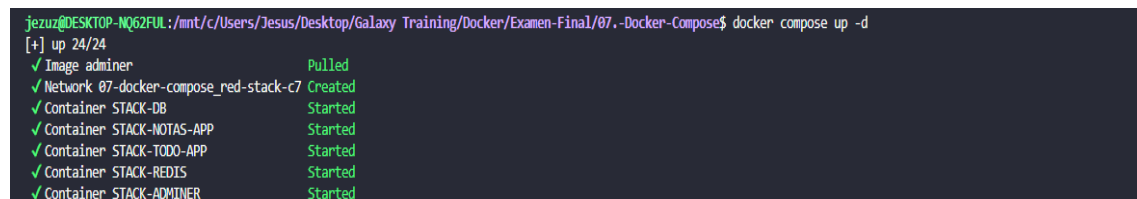


```
07.-Docker-Compose > compose.yml > {} services > {} notas-app > {} networks > abc 0
docker-compose.yml - The Compose specification establishes a standard for the definition of multi-container platform-agnostic applications (compose-spec.json) | >Run All Ser
1  services:
2
3      >Run Service
4      db:
5          image: postgres:16
6          container_name: STACK-DB
7          environment:
8              - POSTGRES_USER=jezuz
9              - POSTGRES_PASSWORD=querty123
10             - POSTGRES_DB=examen_db
11          ports:
12              - "5432:5432"
13          networks:
14              - red-stack-c7
15
16      >Run Service
17      adminer:
18          image: adminer
19          container_name: STACK-ADMINER
20          depends_on:
21              - db
22          ports:
23              - "8080:8080"
24          networks:
25              - red-stack-c7
```

Inicializar los contenedores

Para crear e iniciar los contenedores en segundo plano, ejecutamos:

`docker compose up -d`



```
jezuz@DESKTOP-NQ62FUL: /mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final/07.-Docker-Compose$ docker compose up -d
[+] up 24/24
✓ Image adminer                               Pulled
✓ Network 07-docker-compose_red-stack-c7       Created
✓ Container STACK-DB                           Started
✓ Container STACK-NOTAS-APP                     Started
✓ Container STACK-TODO-APP                     Started
✓ Container STACK-REDIS                         Started
✓ Container STACK-ADMINER                      Started
```

Revisar los contenedores

Para verificar que los contenedores se hayan creado y conocer su estado, utilizamos:

docker compose ps

```
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final/07.-Docker-Compose$ docker compose ps
```

NAME	IMAGE	COMMAND	SERVICE	CREATED	STATUS	PORTS
STACK-ADMINER	adminer	"entrypoint.sh docke..."	adminer	12 seconds ago	Up 10 seconds	0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp
STACK-DB	postgres:16	"docker-entrypoint.s..."	db	12 seconds ago	Up 10 seconds	0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp
STACK-NOTAS-APP	jezuz1904/tarjeta-notas-v2:1.0	"docker-entrypoint.s..."	notas-app	12 seconds ago	Up 10 seconds (healthy)	0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
STACK-REDIS	redis:7	"docker-entrypoint.s..."	cache	12 seconds ago	Up 11 seconds	0.0.0.0:6379->6379/tcp, [::]:6379->6379/tcp
STACK-TODO-APP	jezuz1904/to-do-list-v1:1.0	"unicorn --bind 0.0..."	todo-app	12 seconds ago	Up 11 seconds (healthy)	0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp, 0.0.0.0:5050->5000/tcp, [::]:5050->5000/tcp

Detener los contenedores

Para detener los contenedores que se encuentran en ejecución, ejecutamos:

docker compose stop

```
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final/07.-Docker-Compose$ docker compose stop
```

```
[+] stop 5/5
```

- ✓ Container STACK-TODO-APP Stopped
- ✓ Container STACK-NOTAS-APP Stopped
- ✓ Container STACK-ADMINER Stopped
- ✓ Container STACK-REDIS Stopped
- ✓ Container STACK-DB Stopped

Iniciar nuevamente los contenedores

Si los contenedores fueron detenidos anteriormente, podemos iniciarlos nuevamente con:

docker compose start

```
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final/07.-Docker-Compose$ docker compose start
```

```
[+] start 5/5
```

- ✓ Container STACK-REDIS Started
- ✓ Container STACK-NOTAS-APP Started
- ✓ Container STACK-DB Started
- ✓ Container STACK-TODO-APP Started
- ✓ Container STACK-ADMINER Started

Realizamos las verificaciones

